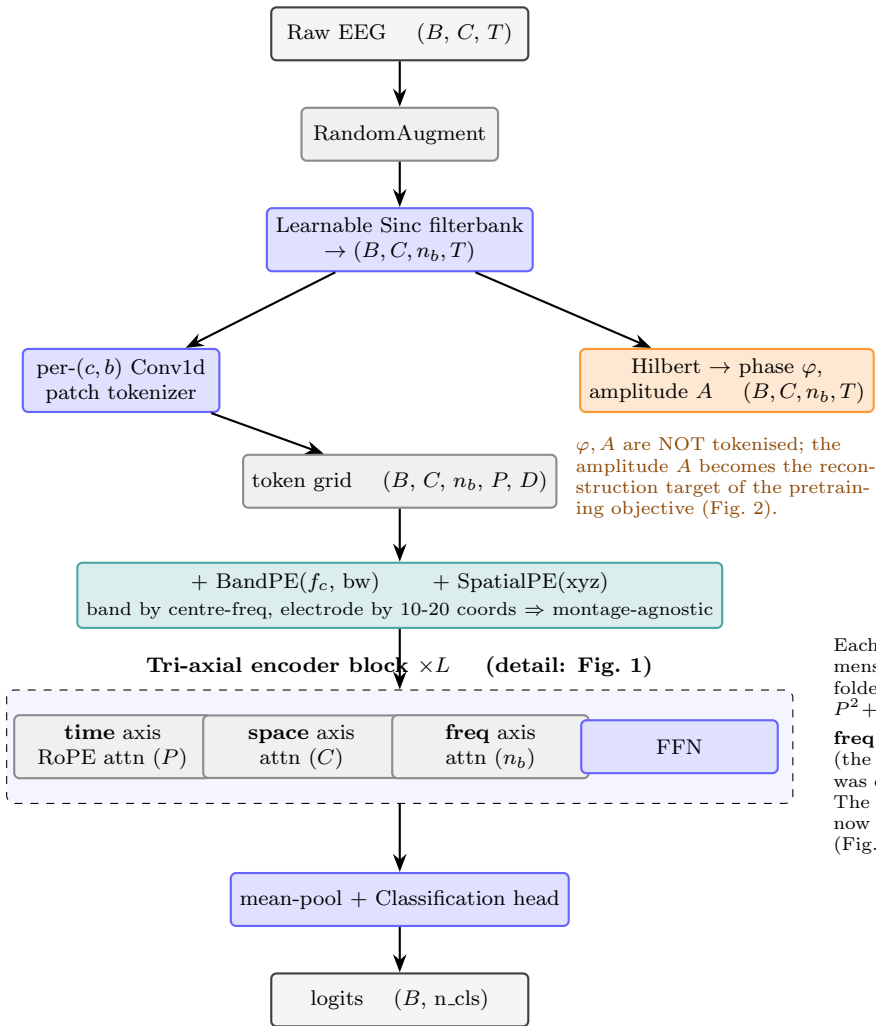


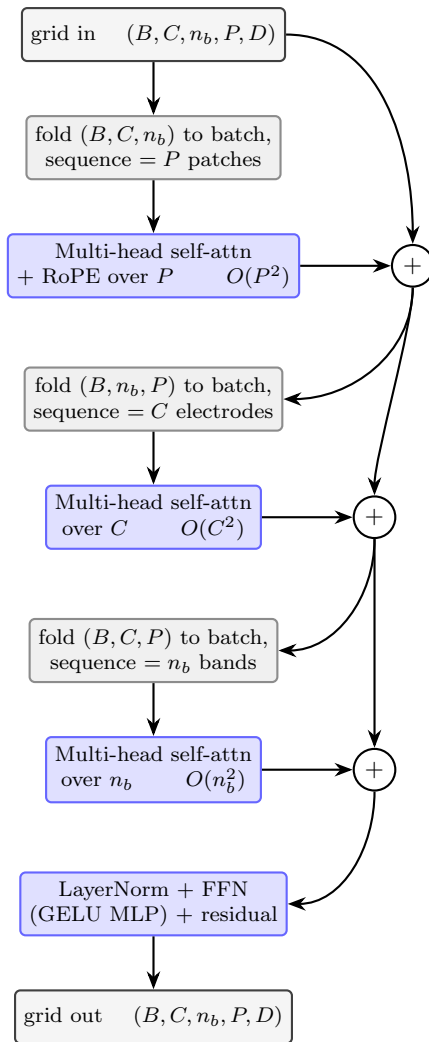
PAC-Former v2 — tri-axial analytic-signal EEG foundation model



Each axis mixes ONE dimension with the other two folded into the batch: cost $P^2 + C^2 + n_b^2$, not $(Cn_bP)^2$.

freq axis = attention
(the v1-v5 PAC *operator* was dropped, sec. 13.20). The frequency PRIOR now lives in the objective (Fig. 2), not the layer.

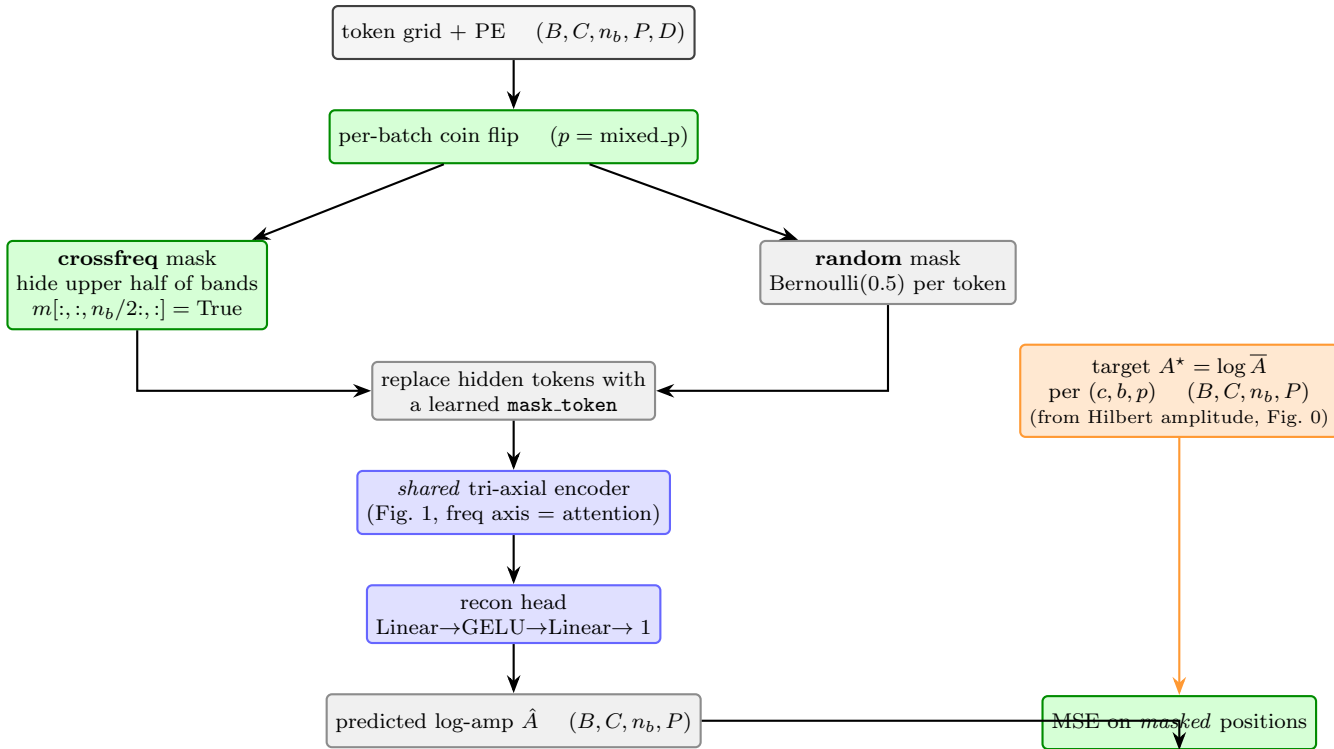
Fig. 1 — Tri-axial encoder block: three factorised axis-attentions (norm + residual each), then FFN



The factorisation is the compute win: the frequency axis is constant in sequence length.

Only the *freq*-axis mixer is swappable in the ablation (attention / coupling / CoTAR); space & time are always attention (sec. 13.8).

Fig. 2 — cf_mixed cross-frequency masked pretraining (sec. 13.21) — the surviving contribution



Why this forces PAC without any operator: the only visible cue to a hidden HIGH-band amplitude is the LOW bands' *phase*, so the one structure that solves the task is low-phase → high-amplitude coupling. Encoding the prior as an *objective* (not a layer) is what lets it survive optimisation (sec. 13.18). **cf.mixed** (coin-flip crossfreq/random) keeps the imbalanced-binary win AND clears standard MAE on multi-class (sec. 13.21).